

Chapter 1. Intro

Partial Differential Equation (PDE).
Functional eq involving multivariable functions and its derivative.

It is not important (at least real variable func.) to study

$$|u_x(x, y)| = -1$$

→ We study typical important PDE.

Example.

$$\Omega \subset \mathbb{R}^n, n \text{ is } 1, 2, 3, \dots$$

$$t \in \mathbb{R}, x = (x_1, x_2, \dots, x_n) \in \Omega \text{ sp. variable}$$

↑
time variable

Example 1.1 < Laplace / Poisson >

For $u: \Omega \rightarrow \mathbb{R}$

$$\Delta u := \frac{\partial^2 u}{\partial x_1^2} + \dots + \frac{\partial^2 u}{\partial x_n^2}$$

is call Laplace operator

For an unknown func.

$$u: \Omega \rightarrow \mathbb{R}$$

$$(4) - \Delta u(x) = 0, x \in \Omega$$

is call Laplace eq

A solution of (4) is called harmonic func.

For given func. $f: \Omega \rightarrow \mathbb{R}$

$$(P) - \Delta u(x) = f(x), x \in \Omega$$

called Poisson eq.

Example 1.2 (Heat / Diffusion eq)

For unknown fun. $u: \Omega \times [0, \infty) \rightarrow \mathbb{R}$

$$(H) \frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, x \in \mathbb{R}^n, t > 0$$

is called heat / Diffusion eq.

Example 1.3 (Wave eq).

For unk func. $u: \Omega \times \mathbb{R} \rightarrow \mathbb{R}$

$$(W) \frac{\partial^2 u}{\partial t^2}(x, t) - \Delta u(x, t) = 0, x \in \mathbb{R}^n, t \in \mathbb{R}$$

is called wave eq.

Example 1.4 (Schrödinger eq).

For unk func. $u: \Omega \times [0, \infty] \rightarrow \mathbb{C}$

$$(S) \frac{\partial u}{\partial t}(x, t) - i \Delta u(x, t) = 0, x \in \mathbb{R}^n, t \in \mathbb{R}$$

is called Schrödinger eq.

Example 1.5 (Navier - Stokes eq)

For unk func. vector-valued func. $\vec{u} = (u_1, u_2, \dots, u_n)$
: $\Omega \times [0, \infty) \rightarrow \mathbb{R}^n$

and unk func. scalar func. $p: \Omega \times [0, \infty) \rightarrow \mathbb{R}$

$$\left\{ \begin{array}{l} \frac{\partial \vec{u}}{\partial t}(x, t) + \vec{u}(x, t) \cdot \nabla \vec{u}(x, t) + \nabla p(x, t) = \vec{0}, x \in \Omega, t > 0 \\ \operatorname{div} \vec{u}(x, t) = 0, x \in \Omega, t > 0 \end{array} \right.$$

is call Euler eq.

$$\begin{cases} \frac{\partial \vec{u}}{\partial t}(\alpha, t) + \vec{u}(\alpha, t) \cdot \nabla \vec{u}(\alpha, t) - \Delta \vec{u}(\alpha, t) + \nabla p(\alpha, t) = \vec{0} \\ \alpha \in \Omega, t > 0 \\ \operatorname{div} \vec{u}(\alpha, t) = 0 \\ \alpha \in \Omega, t > 0. \end{cases}$$

(NS) is called Navier-Stokes eq.

Usually we study PDEs with Initial Data (初期値) and Boundary Data

Main Topic study PDEs.

- 1) existence of solutions
- 2) uniqueness of solutions
- 3) continuity for given data
- 4) study quality and quantity properties

< Review of the vector analysis >

$$\nabla := \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$$

is called **nabla** / **gradient**

② For ~~the~~ vector-valued func.

$$\vec{F} = (F^1, \dots, F^n) : \Omega \rightarrow \mathbb{R}^n$$

$$\operatorname{div} \vec{F} = \frac{\partial F^1}{\partial x_1} + \dots + \frac{\partial F^n}{\partial x_n}$$

is called **divergence**

③ For an scalar func. $f : \Omega \rightarrow \mathbb{R}$

$$\operatorname{grad} f = \nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

is called **gradient**

You (should) know

$$\int_a^b \frac{df(x)}{dx} dx = f(b) - f(a)$$

what is the similar relation for multi-value func?

Thm 1.1 (Gauss's divergence Thm).

Let $\Omega \subset \mathbb{R}^n$ be a b'dd open set with ~~the~~ smooth boundary

Then for any C^1 vector field

$$\vec{F} : \Omega \rightarrow \mathbb{R}^n$$

$$\int_{\Omega} \operatorname{div} \vec{F} d\alpha = \int_{\partial \Omega} \vec{F}(\alpha) \cdot \vec{\nu}(\alpha) d\tau(\alpha)$$

is hold, where $\vec{\nu}(\alpha)$ is the outer unit normal vector at $\alpha \in \partial \Omega$ and $d\tau$ is the volume element of $\partial \Omega$

You should be know

$$\int_a^b \frac{df}{dx}(x) g(x) dx = \left(f(b)g(b) - f(a)g(a) \right) - \int_a^b f(x) \frac{dg(x)}{dx} dx$$

Thm 1.2 (Integration by parts)

Let $\Omega \subset \mathbb{R}^n$ be a b'dd open set with smooth boundary

and let $u, \phi : \Omega \rightarrow \mathbb{R}$ be C^1 -funcs.

Let $\vec{\nu} = (\nu^1(\alpha), \nu^2(\alpha), \dots, \nu^n(\alpha))$ be the outer unit normal vector at $\alpha \in \partial \Omega$. Then

$$\begin{aligned} \int_{\Omega} \frac{\partial u}{\partial x_j}(\alpha) \cdot \phi(\alpha) d\alpha \\ = \int_{\partial \Omega} u(\alpha) \phi(\alpha) \nu^j(\alpha) d\tau(\alpha) \\ - \int_{\Omega} u(\alpha) \frac{\partial \phi}{\partial x_j}(\alpha) d\alpha \end{aligned} \quad (1.3)$$

is hold. (1.3) is called **Integration by parts**